VIT Vidyalankar Institute of Technology
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Department of Computer Engineering

Semester	T.E. Semester V - Computer Engineering
Subject	E-waste
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E-Waste Management: Cross-Country Comparative Report

India · Japan · Russia · USA · Australia

1. Introduction

Electronic waste (e-waste) management approaches vary significantly across countries based on regulatory maturity, industrial capacity, and the role of formal versus informal recycling sectors. This report compares e-waste management frameworks across five countries — India, Japan, Russia, the United States, and Australia — based on their governing approach, legal framework, and system performance.

2. Comparative Overview

Aspect	India	Japan	Russia	USA	Australia
Approach	Extended Producer Responsibility (EPR)	Producer responsibility + strict recycling	Government regulation + producer/importer responsibility	State-based system + federal regulation	Product stewardship
Key Law	E-Waste (Management) Rules, 2022	Home Appliance Recycling Law	Waste management legislation + EPR framework	RCRA + state e-waste laws	Product Stewardship Act 2011
Government Role	Sets rules, registers producers/recyclers, monitors EPR via CPCB	Strong regulation, builds collection/recycling infrastructure	Regulates waste handling and recycling	EPA regulates hazardous waste; states hold major responsibility	Establishes stewardship schemes and standards

Producer Role	Must meet recycling targets, fulfil EPR obligations	Major responsibility for collection and recycling	Growing responsibility for end-of-life products	Varies by state; some take-back obligations	Participate in stewardship programs
Aspect	India	Japan	Russia	USA	Australia
Collection System	Formal system exists; informal sector still significant	Highly organized takeback system	Developing infrastructure	Varies significantly by state	Formal, government/industry-run
Recycling Quality	Growing formal sector; informal recycling remains a challenge	High-tech, efficient	Less developed than Japan	Well-developed but inconsistent across states	Well-established formal programs
Informal Sector	Very significant	Very limited	Present but not central	Relatively limited	Relatively limited
Consumer Role	Use authorized collection/recycling channels	Pay/participate in established take-back systems	Use designated collection systems	Use state/local centres and retailer programs	Return via collection programs
Main Strength	Strong EPR framework, rapidly expanding formal recycling	Very efficient, organized system	Increasing regulatory focus	Flexible state systems, strong private industry	Strong stewardship approach
Main Challenge	Informal recycling, collection gaps, enforcement	High cost, complex processes	Infrastructure and enforcement gaps	Inconsistency across states	Ensuring high collection/recovery rates